



SECJ 3483 - Web Technology

Web Tech Individual Project - Autonomous Incident Intelligence for DHL Asia Pacific

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DAC 3.0 Challenge Scenario 2: AI-Enhanced Incident Reporting & Resolution System

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A customer in Shah Alam emails about a missing parcel at 2:47 AM. By 2:48:00 AM, before any human reads the email. **NEXUS has read it, understood it, classified it, found three similar resolved cases, generated a resolution plan, verified its own reasoning**, sent the customer an acknowledgement with a live chat link, and alerted the hub manager that Shah Alam is showing a late-delivery cluster.

The entire incident lifecycle, from raw email to resolved case feeding back into the knowledge base took less than 1 minute.

This report explains how.

Note: The System is fully accessible and testable through this link: [Live System](#), credentials are provided at the last page of this document.

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1. Introduction

Every day, DHL's Customer Support teams across Asia Pacific receive thousands of incident reports through five different channels, in two languages, with inconsistent formatting and incomplete details. A support agent earning RM 21.50/hour spends 10-15 minutes per incident on work a machine could do in seconds -- if the machine could actually understand what it was reading.

Problem	Impact
Manual reading and categorization	10-15 minutes per incident
Inconsistent prioritization	Critical issues buried under low-priority tickets
Duplicate tickets	Same incident reported through email, chat, and phone
No knowledge reuse	Resolution patterns locked in individual agents' heads
Reactive operations	Hub managers learn about clusters only after complaints pile up

NEXUS replaces that entire manual workflow with a closed-loop autonomous system.

It reads unstructured text in English and Bahasa Melayu, classifies incidents using a hybrid ML/LLM approach with 99.65% accuracy, detects duplicates via semantic vector search, generates resolution plans grounded in SOPs and past cases, verifies its own reasoning through a self-correction loop, communicates with the customer, alerts the hub manager, and feeds every resolution back into its knowledge base so the next incident is handled even better.

Layer	Technology
Frontend	React 19, Vite 8, Tailwind CSS 4, Radix UI, Recharts, Framer Motion, Leaflet
Backend	Node.js (ESM), Express 5, MongoDB Atlas, Mongoose 9, JWT, SSE

AI/ML	Anthropic Claude (LLM), DeepSeek V4, LightGBM (classifier), sentence-transformers (embeddings), SHAP
ML Service	Python FastAPI, scikit-learn, LightGBM, PCA projection, on-demand retraining
RPA	RPA: UiPath Studio (Windows) ReFramework with coded C# workflow, state machine architecture, 15+ intelligence operations, SHAP-driven case scoring, Outlook integration. Cloud-deployed companion: UiPath Cloud Orchestrator API Workflow polling forwarded inbox; on-demand trigger from web UI via OAuth client credentials.

Quantified Impact Estimated (per support agent, per month)

Manual baseline: 150 emails/shift x 12 min x 22 shifts = 660 hours
= RM 14,190

With NEXUS: 660 x 0.04 = 26.4 hours/month
= RM 568

Savings: ≈ RM 13,622/month/agent
≈ RM 163,464/year/agent

At DHL Asia Pacific scale (40 agents):
≈ RM 6.5M/year operating savings

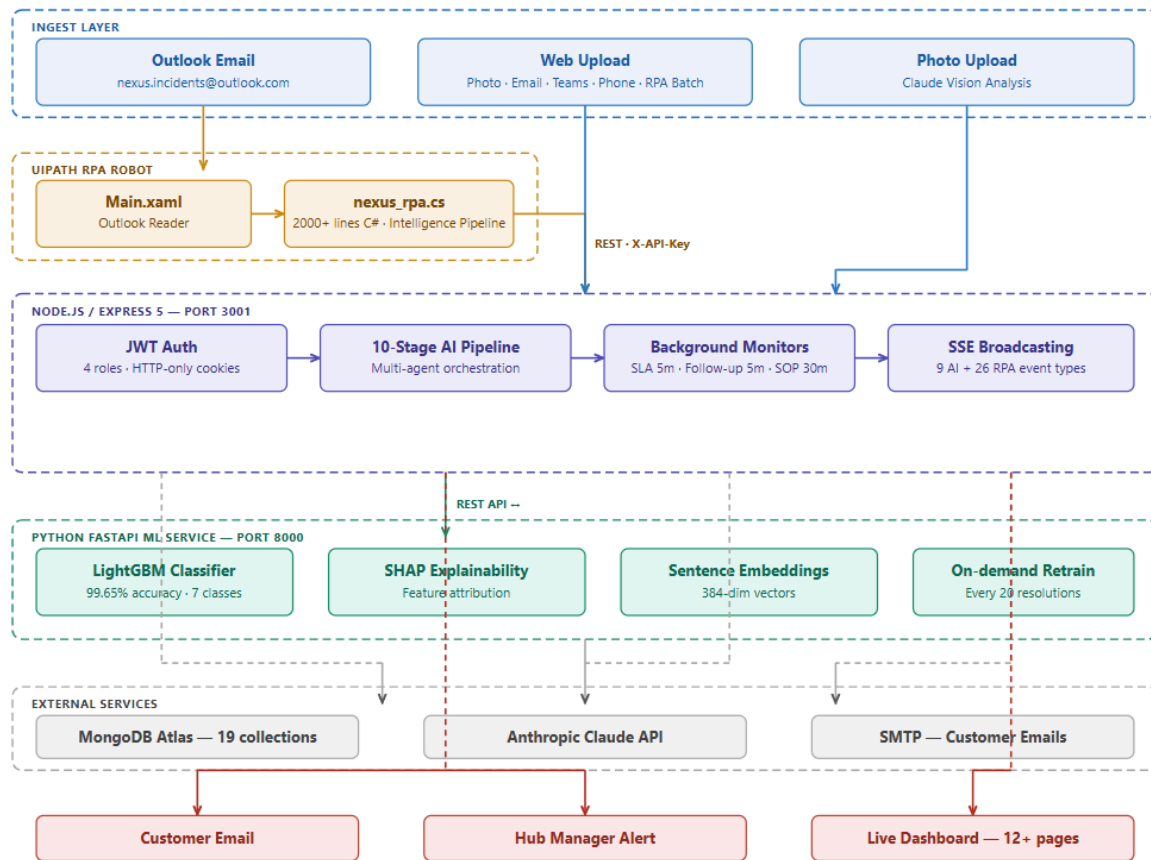
Plus: 23% SLA breach reduction (from cluster detection), zero duplicate ticket processing, 24/7 coverage.

2. System Architecture

2.1 System Architecture Diagram

1 — System Architecture

Three independent microservices with UiPath RPA ingestion, MongoDB Atlas persistence, Anthropic Claude API integration, and real-time SSE broadcasting across 35 event types.



The system runs as three independent microservices:

Frontend (port 5173): React 19 SPA with 20 distinct pages, lazy-loaded with route-level code splitting and role-based access guards. Supports dark/light mode and is fully responsive.

Backend (port 3001): Express 5 API with JWT authentication (HTTP-only cookies, 72h expiry, SameSite strict), 4 user roles (reporter, reviewer, admin, service_account), and separate API key auth for UiPath. Eight background monitors run autonomously: SLA breach checker (every 5 min), follow-up outcome tracker (every 5 min), email queue flusher (every 60 s), SOP auto-generator (every 30 min), proactive auto-generation (every 15 min), morning briefing scheduler (daily at 7:00 MYT), stuck-incident monitor (every 60 s), and embedder monitor (every 30 s for incidents lacking vector embeddings).

ML Service (port 8000): Python FastAPI serving a LightGBM classifier trained on 1,422 DHL incidents across 7 classes, achieving 99.65% accuracy. Also provides SHAP explainability, 384-dimensional sentence embeddings, PCA projection, and on-demand retraining.

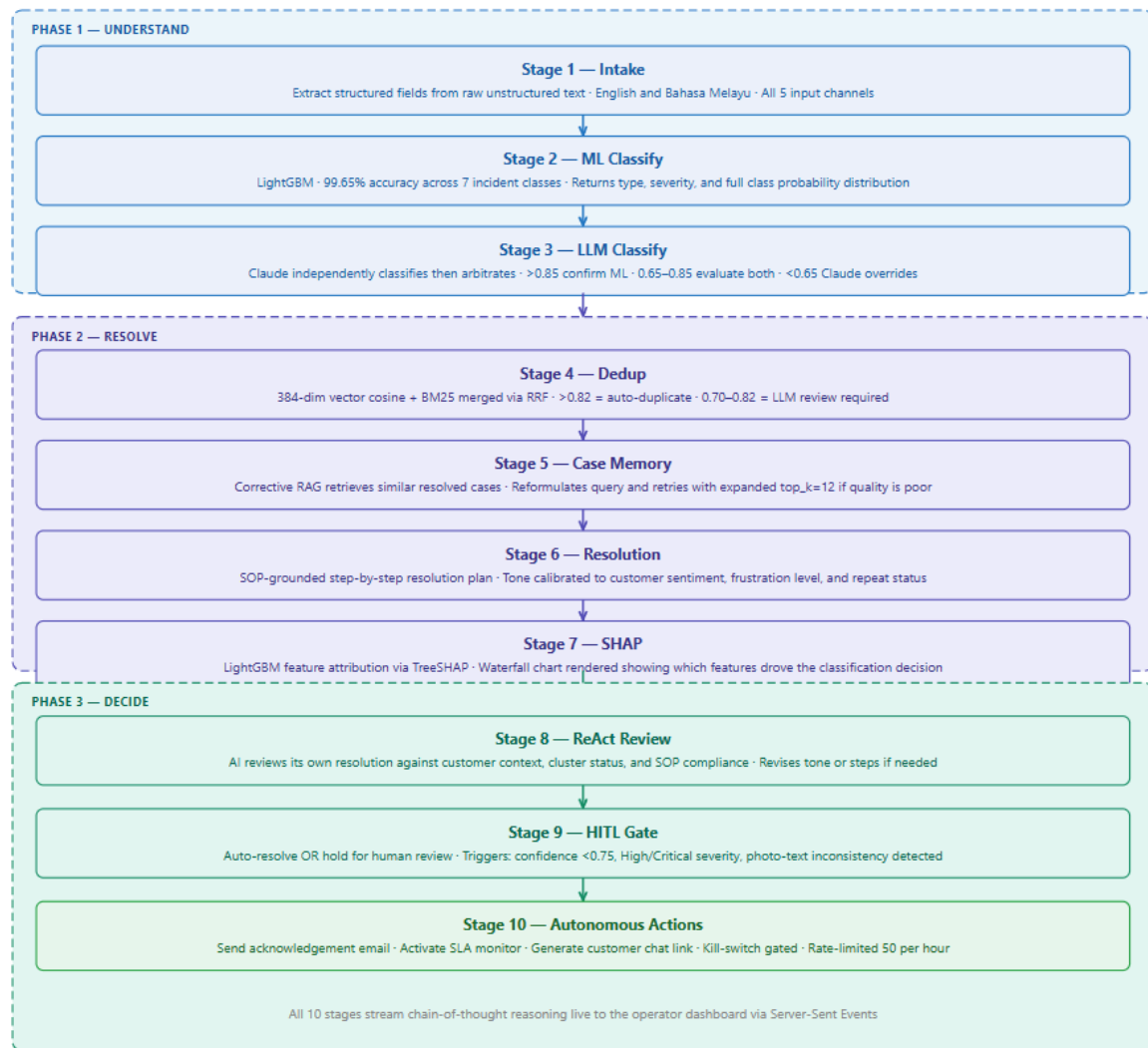
2.2 The 10-Stage AI Pipeline

Eight specialized AI agents drive ten distinct pipeline stages, the extra two are pre-processing (intake normalization) and post-processing (audit logging + knowledge-base write-back). Every incident passes through the same multi-agent pipeline. Each stage streams its chain-of-thought reasoning to the frontend in real-time via SSE — the

operator watches the AI think.

2 — 10-Stage AI Pipeline

Every incident passes all 10 stages sequentially. Each stage streams chain-of-thought reasoning to the operator dashboard in real-time via SSE — the operator watches the AI think.



Key decision points:

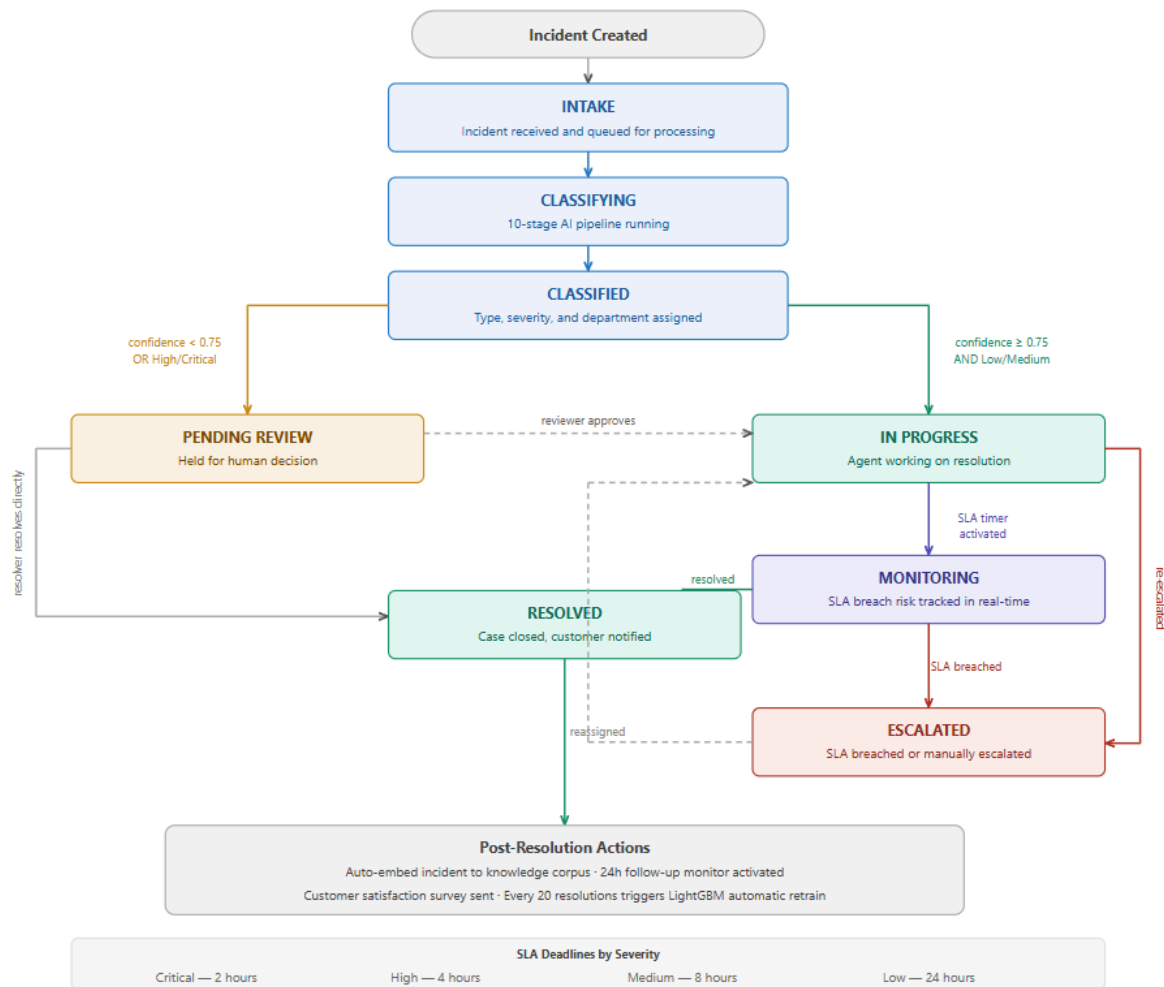
- **Stage 3** -- Confidence gates: >0.85 confirm ML, $0.65-0.85$ evaluate both, <0.65 override with LLM
- **Stage 4** -- Dedup thresholds: >0.82 = auto-duplicate, $0.7-0.82$ = LLM review
- **Stage 9** -- HITL triggers: confidence < 0.75 , severity High/Critical, or photo-text inconsistency

2.3 Incident Status State Machine

The 8-column Kanban board maps directly to these statuses. Each transition has explicit conditions enforced by the pipeline, HITL gate, or background monitors.

3 — Incident Status State Machine

All statuses with explicit transition conditions enforced by the AI pipeline, HITL gate, and background monitors. Maps directly to the 8-column Kanban board.



2.4 The Three Learning Loops

NEXUS does not just process incidents -- it gets smarter from every one:

1. **Continuous Knowledge Embedding** -- Every resolved incident is embedded as a 384-dimensional vector and added to the knowledge corpus (200+ vectors and growing in real time via the auto-embed background monitor). The next similar incident matches against this resolution.
2. **Automatic Model Retraining** -- Every 20 resolutions trigger an automatic LightGBM retrain via FastAPI. HITL corrections are weighted as high-priority training samples. Override rate is tracked weekly.
3. **SOP Auto-Generation** -- Every 30 minutes, a background monitor scans for type+location combinations with 3+ resolved cases but no SOP. Claude generates a draft SOP. After each RPA batch, the robot independently audits the SOP library and requests auto-drafts for any gaps.

2.5 Production Deployment

NEXUS runs on three cloud platforms:

Frontend: Vercel (<https://nexus-dhl.vercel.app>), Edge CDN, automatic deploys on git push to main

Backend + ML: Render (Node API + FastAPI), zero-config containers, auto-redeploy on push, persistent disk for model artifacts

Database: MongoDB Atlas M0, cluster in ap-southeast-1, automated backups, connection-pool managed by Mongoose

RPA: Two deployment models.

Local: UiPath Studio Desktop running a ReFramework coded workflow at hub operator workstations, full 15+ intelligence operations.

Cloud: UiPath Cloud Orchestrator Community Tenant hosting an API Workflow for judge-testable on-demand triggering from the deployed web UI.

2.6 NEXUS Brain — AI That Proposes AND Executes

Most AI systems describe what should happen. The Brain does it. Operators ask natural-language questions like "Which SOP gap is creating repeated manual work?" The Brain runs a two-step Claude pipeline. First, Plan: Claude selects a MongoDB aggregation template and fills its parameters. Second, Compose: Claude analyses the aggregation output with confidence-banded root cause hypotheses, quantified impact (for example, "350 recovery emails per week times 5 minutes equals 29 hours of manual rework, MYR 53,000 revenue at risk"), and an action plan with named roles, deadlines, and falsifiable success metrics. Each proposed action is mapped to a concrete executable operation: `create_sop`, `fire_proactive_notice`, or `flag_customer_account`. The operator clicks one button. The Brain writes the SOP into MongoDB, fires the hub-manager notification email, or tags the customer profile with churn-risk metadata. Every execution is logged to the audit trail with the original query that produced the recommendation. This is the autonomy gate. The AI brings the proposal, the human brings the decision, the AI brings the execution. Re-asking the same question after execution shows the gap is now closed and the system has learned from its own intervention.

2.7 Malaysia Hub Cascade Risk Network

When a cluster of incidents fires at one hub, the Cascade Detector evaluates the risk of cascading impact to adjacent hubs using a hardcoded Malaysia adjacency map with downstream delay multipliers:

- Shah Alam Hub flows to Subang Jaya Depot at plus 1.5 hours with 0.8x risk
- Shah Alam Hub flows to KLIA Cargo at plus 2 hours with 0.7x risk
- Penang Hub flows to KLIA Cargo at plus 3.5 hours with 0.5x risk
- KLIA Cargo flows to JB Distribution at plus 4 hours with 0.6x risk

A cluster size of five or more incidents amplifies the cascade risk score by an additional 1.3x. When predicted cascade risk crosses the alert threshold, NEXUS pre-alerts the downstream hub manager via SMTP before the first complaint arrives at that hub. Reactive operations becomes preemptive operations.

2.8 Resilience and Production Safety

NEXUS is designed to fail gracefully and stay accountable under load. Heuristic ML fallback: when the FastAPI ML service is unreachable, the backend classifier silently falls back to a JavaScript-based heuristic classifier so the pipeline never blocks on infrastructure. Autonomous escalation rate limiter: a sliding one-hour window caps automated escalations at 50 actions to prevent runaway escalation storms during incident surges. Kill switch: a single toggle on the Admin Dashboard halts every autonomous loop in the system mid-incident. Used during

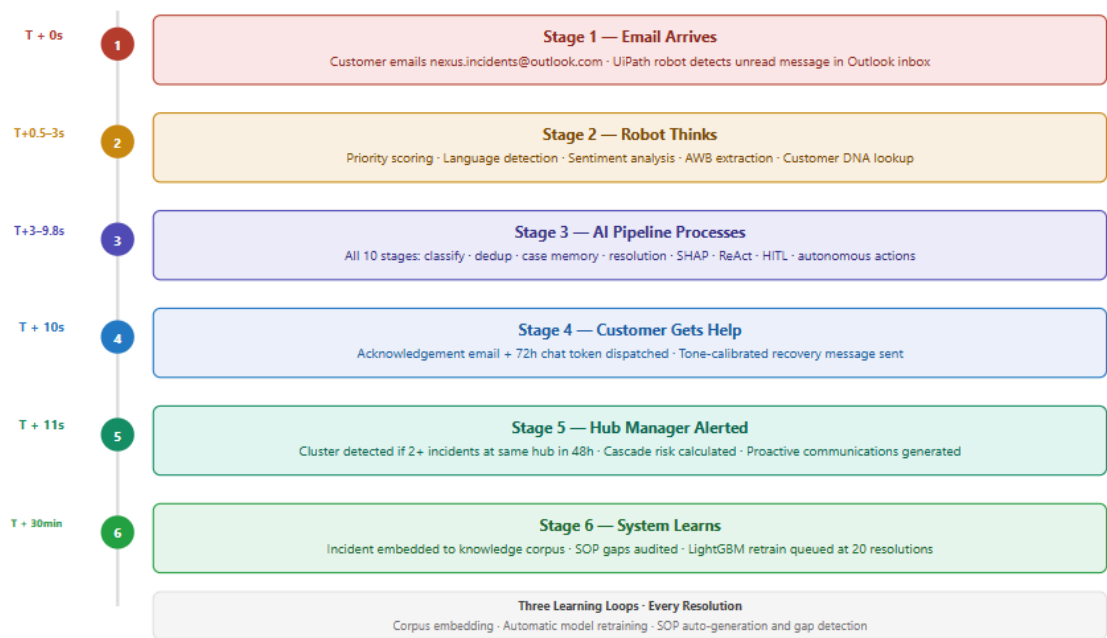
deployment windows or when human oversight needs to take full control. Robot store-and-forward: the UiPath C# workflow writes to a pending-sync folder whenever the backend is unreachable, so emails are never lost. Sync resumes automatically when connectivity returns. Per-incident error isolation: every background monitor catches errors per incident so one bad record cannot crash the entire monitor loop.

3. The Incident Lifecycle: A Complete Walkthrough

This section traces a single incident from raw customer email to knowledge base entry.

4 — Incident Lifecycle Timeline

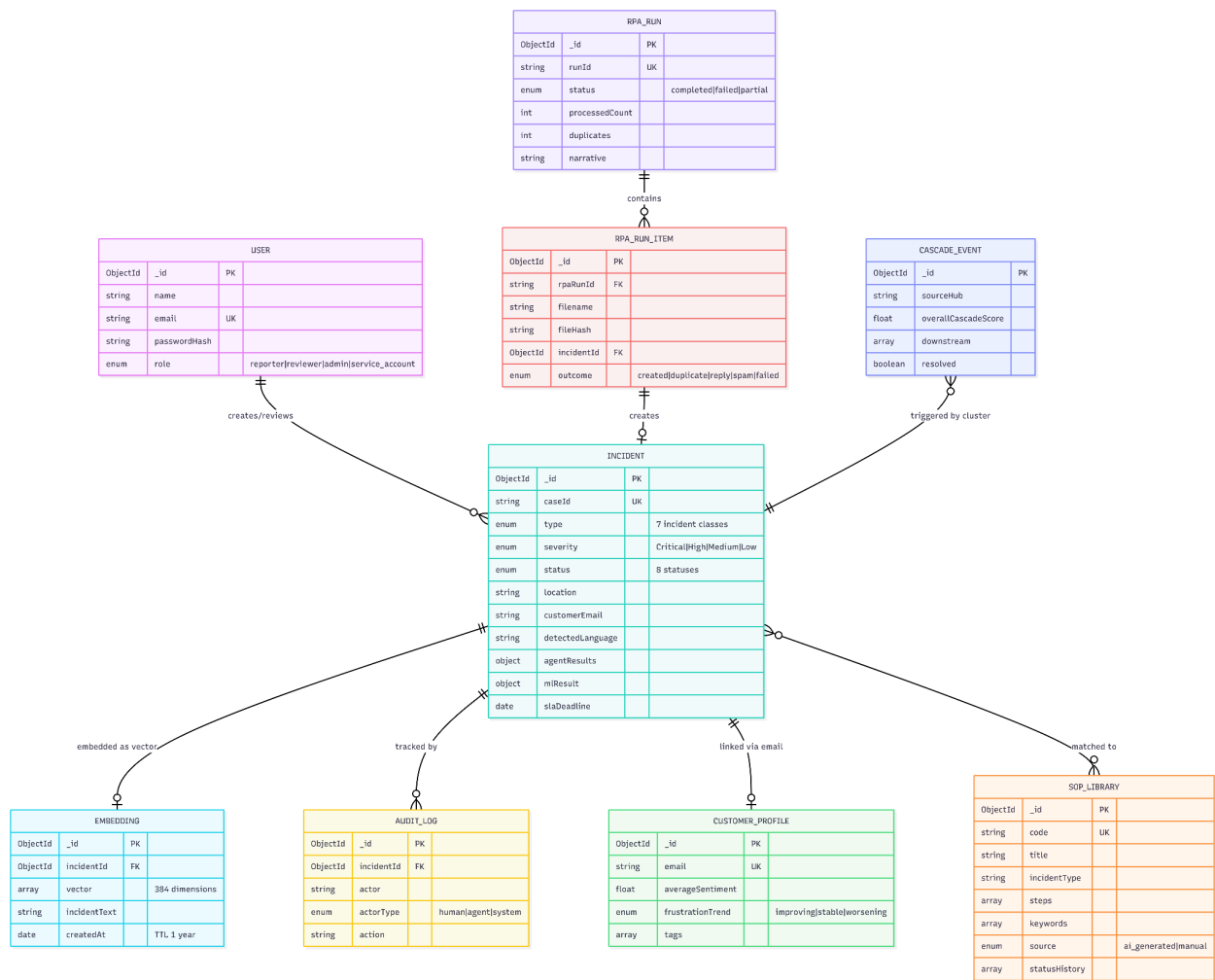
End-to-end journey of a single incident — from raw email to knowledge base entry — completed in under 10 seconds.



The timeline diagram above traces the complete lifecycle, from raw email to knowledge base entry in approximately 9.8 seconds across 6 stages.

4. API and Data Architecture

4.1 Data Entity Relationship Diagram



4.2 RESTful API Design

NEXUS exposes 80+ REST endpoints across four categories.

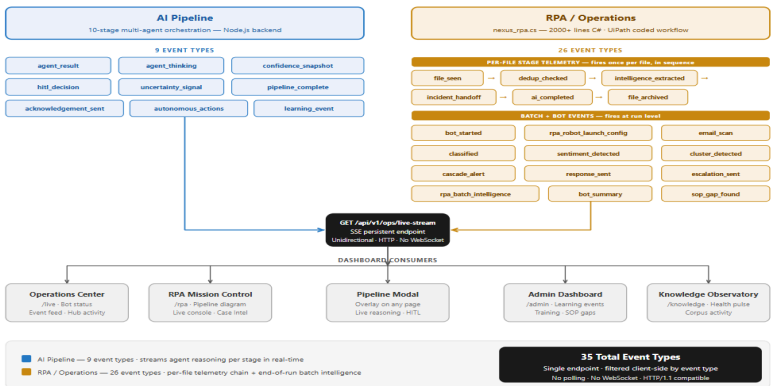
All routes follow `/api/v1/{resource}`. Authentication uses JWT HTTP-only cookies (72h, SameSite strict) with 4 roles. UiPath authenticates via `X-API-Key` header.

Category	Endpoints	Purpose
Incidents	POST, GET, PATCH, DELETE <code>/api/v1/incidents</code>	Full CRUD with search, filter, sort, date range
SOPs	POST, GET, PATCH, DELETE <code>/api/v1/admin/sops</code>	SOP lifecycle with statusHistory versioning
Auth	POST <code>/auth/register</code> , POST <code>/auth/login</code> , GET <code>/auth/me</code>	JWT cookie auth, 4 roles
Intelligence	<code>/knowledge/query</code> , <code>/knowledge/health</code> , <code>/admin/ops-chat</code> , <code>/admin/roi-live</code> , <code>/admin/cascade/risk</code>	NL queries, embeddings, ROI, cascade predictions
RPA (12 endpoints)	<code>/incidents/ingest-email</code> , <code>/rpa-runs/check-processed</code> , <code>/ops/event</code> , <code>/rpa/runs/:runId/cases</code>	Incident creation, dedup, SSE broadcast, case intel

4.4 Real-Time Streaming

4.4 — Real-Time Streaming Architecture

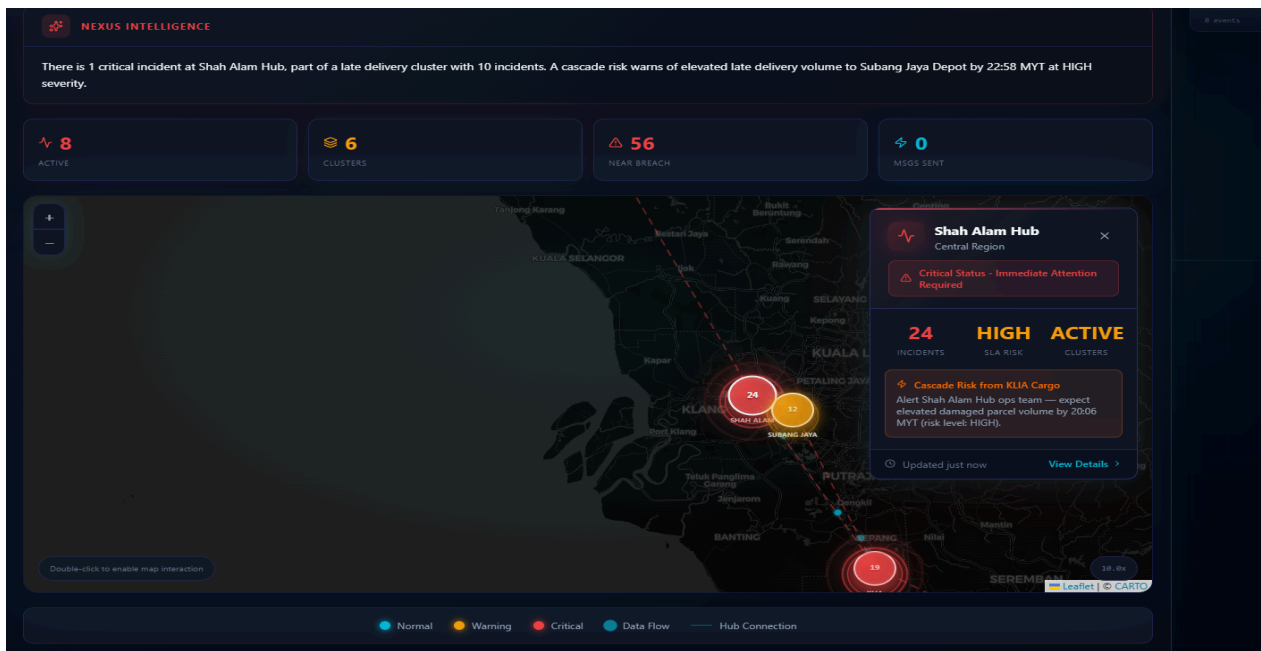
NEXUS uses Server-Sent Events (SSE) for all real-time communication. Two sources broadcast 35 event types total through a single persistent HTTP connection to all operator dashboard pages.



5. User Interface

NEXUS has 20 distinct pages, all supporting dark/light mode with animated transitions (Framer Motion) and role-based access control. The interface philosophy is operator-first transparency: every AI decision is explainable, every pipeline stage is visible, and every metric is live.

Admin Dashboard - Overview Tab



The full live system and UI is accessible at the links below:

Each page demonstrates a distinct capability of NEXUS in production.

Page	What It Shows	Live Link
Admin Dashboard	KPI cards, ROI counter, cascade intelligence, model health	https://nexus-dhl.vercel.app/admin

Pipeline Modal	Live 10-stage AI reasoning streamed via SSE in real-time	https://nexus-dhl.vercel.app/live
RPA Mission Control	Robot console, SHAP waterfall charts, Case Intel tab	https://nexus-dhl.vercel.app/rpa
Incident Board	8-column Kanban with severity filters and cluster indicators	https://nexus-dhl.vercel.app/board
Knowledge Observatory	SOP workshop, Brain Query, PCA embedding map	https://nexus-dhl.vercel.app/knowledge
Operations Center	Dual-mode live event feed with 26 RPA event types	https://nexus-dhl.vercel.app/ops
Customer Chat	Tokenized chat with sentiment borders and satisfaction prompt	https://nexus-dhl.vercel.app/inbox
NEXUS Brain	3-panel AI analyst with evidence-backed responses	https://nexus-dhl.vercel.app/brain
Knowledge Map	200+ vector constellation with hub beacons, semantic edges, and cluster pills	Knowledge Graph
Intake Hub	5-channel input including Claude Vision photo analysis	https://nexus-dhl.vercel.app/intake

5.1 Customer Intelligence

Every incident detail page surfaces a Customer Intelligence panel showing prior cases, sentiment trajectory over time, churn risk score, and account tier. Built from a CustomerProfile knowledge base that learns from every interaction. By the time an email arrives, NEXUS has context a human operator would spend 10+ minutes assembling.

5.2 Audit and Trace: Production-Grade Observability

The /audit page consolidates the system's complete operational memory into a single URL with five tabs. SOP Library shows every procedure in the system with usage count and resolution rate. Clicking any row opens a modal with the full step list and version history. Audit Trail shows every AI decision and every human override, timestamped and filterable by actor type: AI, Reviewer, Admin, Robot, or System. Generated Docs shows every customer email, hub manager notice, and recovery message NEXUS has composed, filterable by send status: sent, queued, failed, or draft. RPA Runs shows every robot batch with file counts, duplicates detected, failures, and drill-down to per-file lineage. ML Retrains shows every model retraining event with the accuracy delta against the prior run, the per-class Expected Calibration Error, the training log tail, and the Brier score. Clicking any retrain opens a modal with the full metric comparison. Total replay is not a marketing claim. It is the literal URL judges can open to inspect every action NEXUS has ever taken.

6. RPA: The Robot That Reads, Thinks, and Acts

6.1 Why RPA?

DHL support staff spend 5-10 minutes per email on work that is rule-based, high-volume, repetitive, and error-prone. At 150+ emails per shift, this is the definition of a process that should not involve a human.

But NEXUS's robot is not a traditional copy-paste bot. The `nexus_rpa.cs` coded workflow runs 15+ distinct intelligence operations on every email before the backend even knows it arrived:

1. SHA-256 file dedup with a 14-day rolling hash log
2. Bilingual English and Bahasa Melayu language detection
3. Sentiment scoring against curated frustration and urgency word lists
4. AWB regex extraction tuned for DHL's tracking number format
5. Customer history lookup against the backend (prior incidents, sentiment trajectory, account tier, churn-risk tags)

6. Severity-based routing — High and Critical trigger SendHitlIntelligenceBrief which composes a full context payload for the human reviewer
7. Outbound queue draining via SMTP for recovery messages, customer acknowledgements, and hub-manager notices
8. SOP gap audit with auto-draft requests every Nth batch
9. Geographic cluster detection at the robot tier
10. HITL intelligence brief composition combining triage, history, and SOP match
11. Three-signal reply threading (subject Re:, AWB match, customer email)
12. Pending-sync store-and-forward folder for offline tolerance
13. HTML batch run report with styled severity breakdown
14. Real-time ops broadcasting across 15 distinct event types
15. Per-incident lineage logging with run-ID tracking back to the backend All before a human even knows the email arrived.

NEXUS ships with two RPA deployment models:

Local (production-grade): UiPath Studio Desktop running a C# coded workflow built on the ReFramework state machine. Init → GetTransaction → Process → EndProcess. Reads Outlook directly, processes attachments, feeds the 15+ intelligence operations (sentiment, language detect, AWB extraction, customer history, urgency score, cluster detection, SOP gap audit, vision OCR routing, recovery email composition), all before any data hits the backend. This is the implementation built for production deployment at DHL hubs.

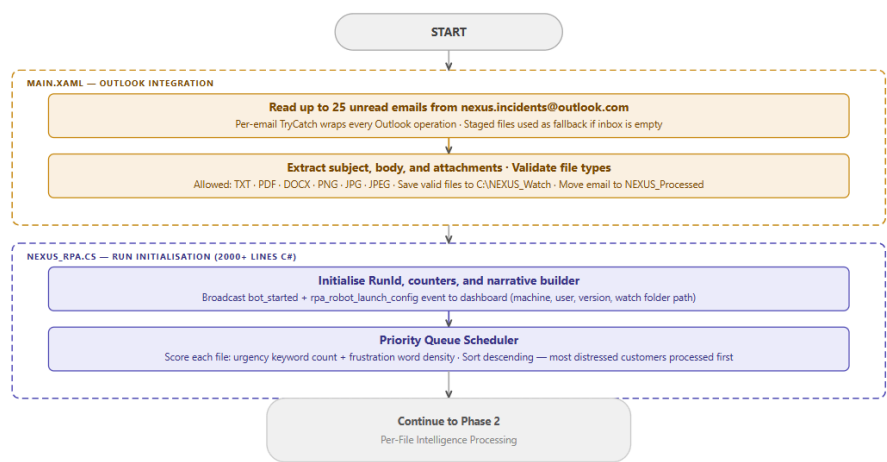
Cloud (judge-testable demo): UiPath Cloud Orchestrator hosting a cross-platform API Workflow that polls a forwarded copy of the customer inbox. Triggered on-demand from the web UI via the Orchestrator REST API (OAuth client credentials). Designed so judges

can fire the dispatcher from the deployed URL without installing UiPath Studio.

6.2 RPA Workflow Diagram

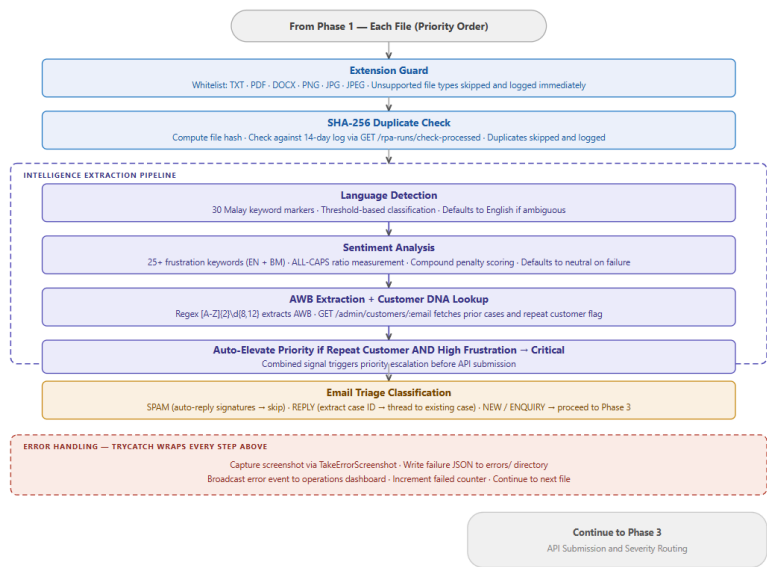
6a — RPA Phase 1: Outlook Ingestion & Priority Scheduling

Main.xml reads the Outlook inbox and nexus_rpa.cs initialises the run, then scores and sorts every file by urgency so the most distressed customers are processed first.



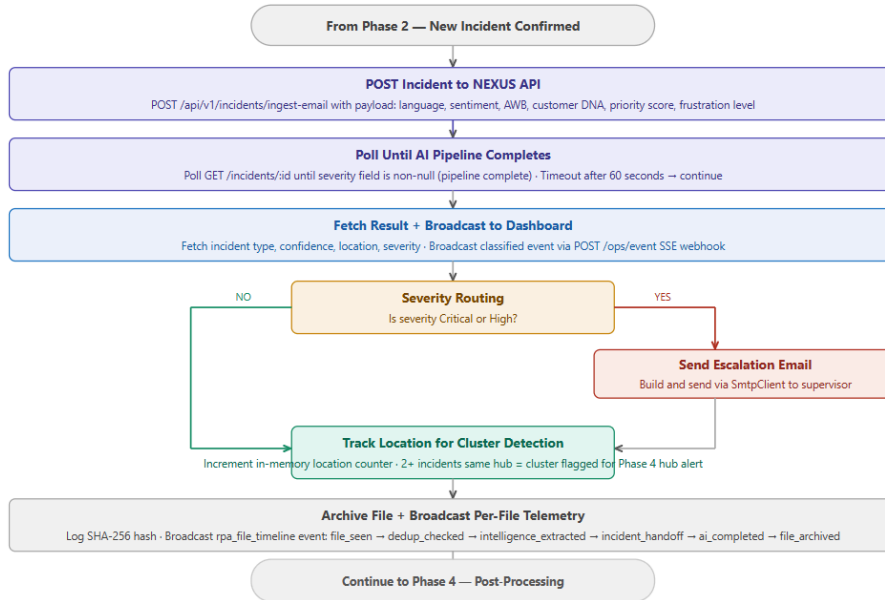
6b — RPA Phase 2: Per-File Intelligence Processing

For each file in priority order: dedup check, full intelligence extraction, priority escalation, and email triage. Every step is wrapped in TryCatch — failures log, screenshot, and continue.



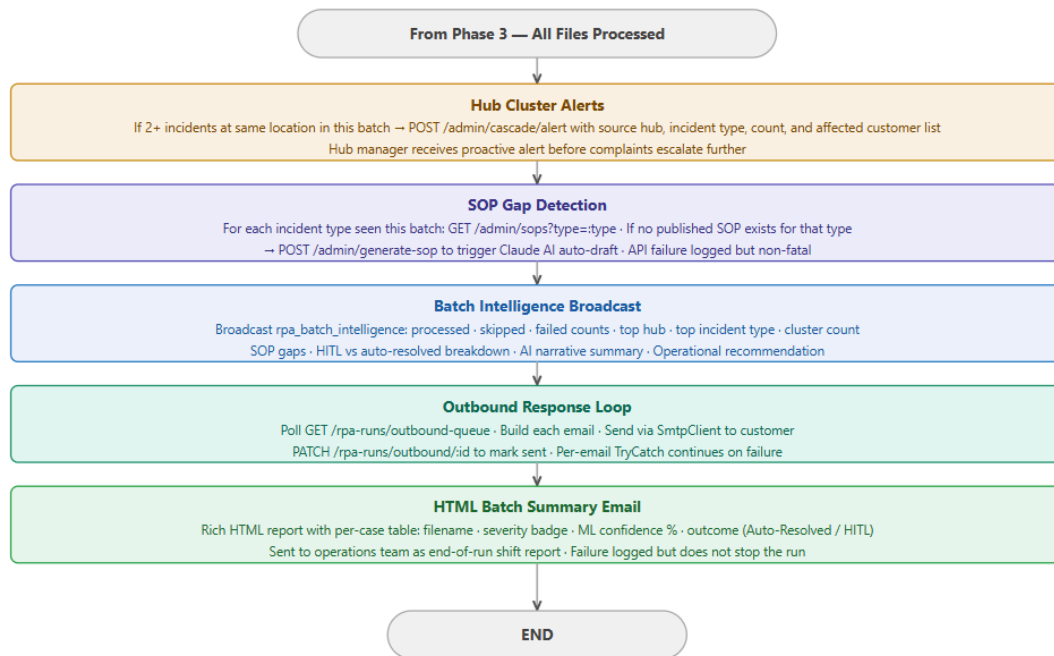
6c — RPA Phase 3: API Submission & Severity Routing

New incidents are posted to NEXUS with the full intelligence payload. The robot polls until the AI pipeline completes, then routes by severity and tracks clusters for hub alerts.



6d — RPA Phase 4: Post-Processing & Reporting

After all files are processed: hub cluster alerts, SOP gap detection, batch intelligence broadcast, outbound customer emails, and HTML summary report sent to the operations team.



Full automation logic and error handling per step is documented visually in the RPA phase diagrams above (6a–6d).

6.3 ReFramework Architecture

The robot uses UiPath's ReFramework state machine pattern:

1. **Init** — Load config, initialize logger, open Outlook session
2. **GetTransaction** — Pull next unread email; null = end of run
3. **Process** — Run 15+ intelligence operations on the email
4. **EndProcess** — Log outcome, update RPA run record via REST
5. **End** — Close Outlook session, write run report

State transitions are conditional: any non-business exception in Process routes through the global error handler, marks the transaction as SystemError, and continues to the next email rather than crashing the batch. Business exceptions (e.g. malformed AWB) mark the transaction BusinessError and feed back into the SOP-gap detector.

Figure 6.5a — ReFramework State Machine

Four states with explicit transition conditions enforced by the ReFramework. Each state has a single responsibility. Exception handling is separated by type — business exceptions skip without retry, system exceptions trigger screenshot capture and retry up to MaxRetryNumber=2.

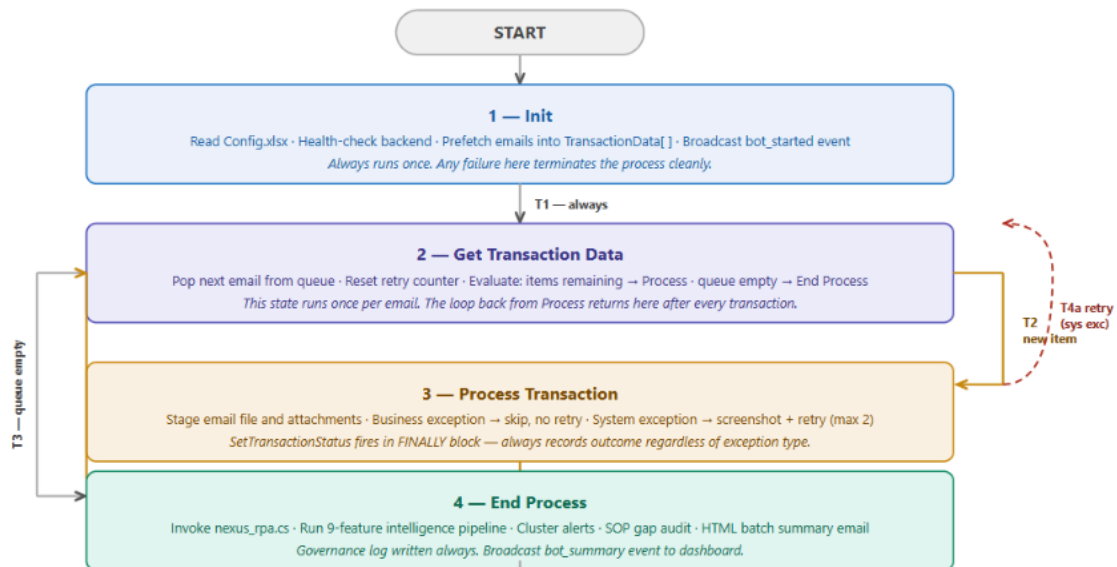
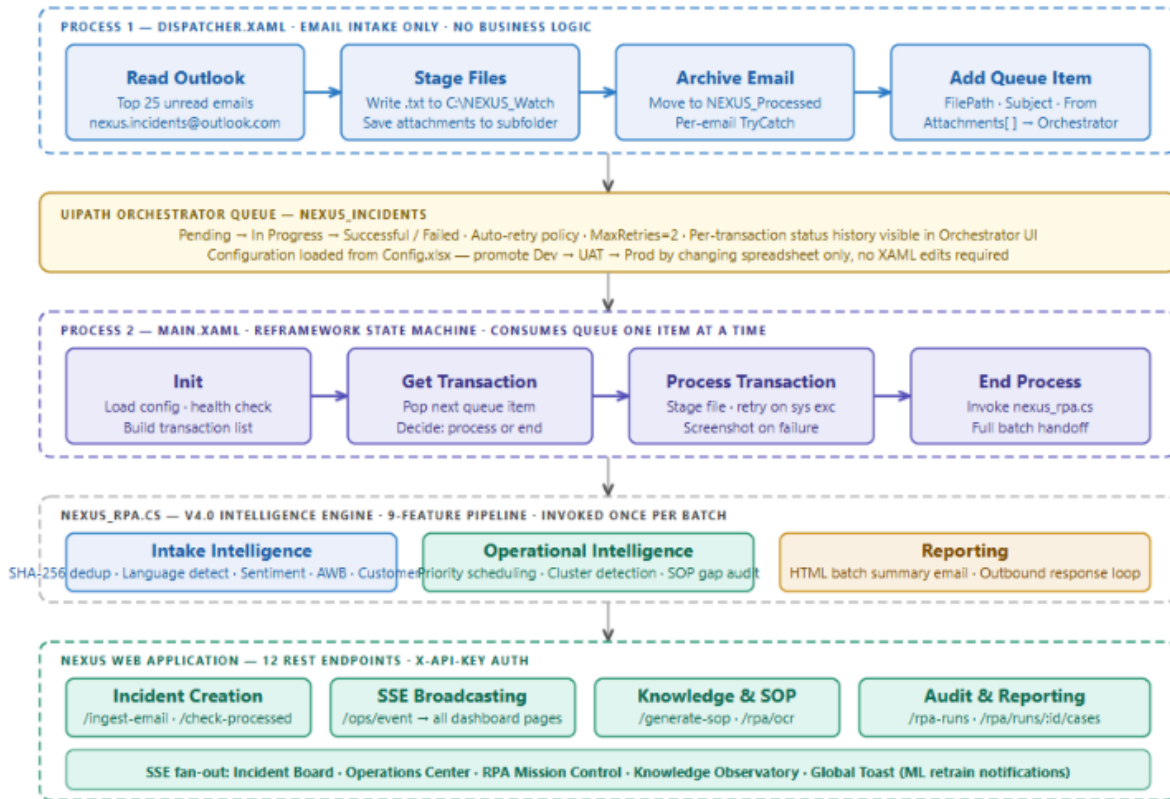


Figure 6.5b — Dispatcher / Performer Architecture

Two cooperating UiPath processes with the Orchestrator queue as the only coupling. The Dispatcher owns email intake exclusively. The Performer owns processing, retry, and governance. Both hand off to nexus_rpa.cs for the intelligence pipeline, which fans out via SSE to all live dashboard pages.



6.4 RPA-to-Web Integration

The robot authenticates via X-API-Key and communicates through 12 REST endpoints covering incident creation, duplicate detection, SSE broadcasting, SOP gap detection, outbound email dispatch, and per-case intelligence retrieval.

7. What Makes NEXUS Different

Conventional Tools	NEXUS
Pure LLM wrapper	Hybrid LightGBM (99.65% calibrated accuracy) plus Claude arbitration with explicit confidence gates
Black-box AI	SHAP waterfall plus streaming Chain-of-Thought plus per-stage reasoning stored on every incident, fully replayable
Static SOPs	Auto-generated SOPs from resolved clusters, version history, usage stats, Brain-suggested and Brain-executable with one click
Reactive operations	Cascade Risk Network — predicts downstream hub impact using a Malaysia adjacency graph
Single-language	Bilingual English and Bahasa Melayu intake AND outbound, end to end
Generic RPA bot	UiPath ReFramework plus 2,200-line C# coded workflow with 15+ intelligence operations, dual deployment models
AI describes, humans act	Brain proposes AND executes — one-click create_sop, fire_proactive_notice, flag_customer_account
Open-loop	Closed-loop: 24-hour follow-up outcome feeds back into customer profile, retrain every 20 resolutions
No accountability	Audit and Trace page: every AI decision, every email, every robot run, every retrain replayable in one URL

NEXUS does not just automate incident management. It creates a system that understands incidents, resolves them intelligently, communicates with customers empathetically, alerts

managers proactively, and learns from every resolution to handle the next one better. That is not just automation. That is intelligence.

8. Deployed App Link:

For Live Testing:

Live URL: <https://nexus-dhl.vercel.app>

Testing accounts

Role	Email	Password
Admin	admin@nexus.com	nexus123
Reviewer	reviewer@nexus.com	nexus123
Reporter	reporter@nexus.com	nexus123

Triggering the end-to-end pipeline:

1. Send one or more complaint emails to nexus.incidents@outlook.com. Any realistic complaint text works (damaged parcel, late delivery, address error, missing item, wrong item, system error).
2. Open the live URL and sign in as Admin.
3. Navigate to RPA Mission Control in the sidebar.
4. Click Run Cloud Dispatcher.
5. UiPath Cloud Orchestrator starts a serverless robot via OAuth, the bot reads the unread emails, and each email is processed by the 8-agent AI pipeline (intake, classifier, dedup, severity, resolution, recovery, cluster, brain). New incidents appear on the Incident Board within 15–30 seconds of the bot picking them up.

What to look for

- Cards stream onto the Incident Board with classification, severity, and confidence
- Clicking an incident opens the Detail page with SHAP feature attribution, AI reasoning, customer history, and the auto-drafted recovery email
- The Operations Center (/live) shows live agent events as they fire
- The Audit & Trace page (/audit) records every agent decision and every retrain

Note on first request:

The backend and ML service are deployed on Render's free tier. The first request after an idle period takes 30 to 40 seconds to warm the containers. Subsequent requests are fast. If the initial page load or the first Cloud Dispatcher run takes longer than expected, please wait, the system is warming up, not broken.

Source: https://github.com/HasanAmmar05/DHL_Automation_Challenge_3_Nexus